Pratush Rai

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RESEARCH EXPERIENCE

Dr. Jiang's Lab, San Jose State University **July 2026 - Present**
Robotics Simulation & Policy Research Assistant *San Jose, CA*

- Engineered a high-fidelity digital twin of the Franka Research 3 and Realhand L6 in NVIDIA Isaac Sim, integrating kinematic, actuator, and contact models to enable physics-based robotic manipulation benchmarking.
- Developed a scalable simulation pipeline using NVIDIA Isaac Lab and PyTorch, enabling parallelized training across 128 simulated environments and reducing dataset generation time by 60%.
- Processed synchronized teleoperation datasets to identify and fit simulation parameters such as friction, compliance, and actuation delay, reducing the sim-to-real gap for contact-rich tasks.
- Supported model-based reinforcement learning in Isaac Lab using imitation-learned priors from teleoperation data and calibrated domain randomization for sim-to-real policy transfer.

Dr. Jiang's Lab, San Jose State University **May 2025 - June 2026**
Wearable Robotics & AI Control Research Assistant *San Jose, CA*

- Developed an end-to-end EMG-to-assist control pipeline for the Kenqing Ant-C1-E hip exoskeleton, mapping surface EMG intent signals and IMU gait features to real-time, phase-synchronized torque actuation.
- Constructed subject-specific musculoskeletal digital twins using MATLAB to simulate biological hip joint moments and establish optimal control targets for stance-to-swing transition assistance.
- Facilitated the diagnosis of robot power failure through multimeters and oscilloscopes to pinpoint error zones; replaced parts based on diagnosis and maintained the robot accordingly.

Bristol Myers Squibb **June 2025 - August 2025**
Product Development Intern *New Brunswick, NJ*

- Enhanced a site-wide GxP raw materials dashboard by integrating automated workflows, improving document accessibility, testing visibility, and cross-functional collaboration for over 500 employees.
- Designed and implemented a centralized workflow tracking system integrated into the site dashboard, providing real-time visibility into raw material testing progress, improving productivity by 20%.
- Led the development of a reference standard library of a new generation spectrometer & osmometer for raw material identification in pharmaceutical testing, used by the ASO team.

MEDFREE Inc. **May 2024 - December 2024**
Research & Development Engineering Intern *Newark, CA*

- Developed design, process, and use-related Failure Mode and Effects Analyses (dFMEA, pFMEA, and uFMEA) to identify and mitigate potential risks associated with a transcatheter valve repair system.
- Utilized SolidWorks to refine the device handle, implementing iterative improvements based on testing feedback from clinical specialists.
- Led prototype manufacturing and testing activities while supporting design control documentation, quality management processes, and regulatory compliance.

Juniper Networks **May 2024 - August 2024**
Hardware Engineering Intern *Sunnyvale, CA*

- Integrated an 800 Gbps networking platform with Juniper Paragon Insights Controller, developing automated hardware monitoring scripts for critical chassis components and improving testing productivity by 40%.
- Conducted extensive experiments to identify correlations between various statistics and device health. Selected key metrics for monitoring, enhancing system observability and reliability.
- Authored a comprehensive user manual detailing the implementation and utilization of Paragon Insights Controller, facilitating onboarding, and ensuring continuity in tool adoption.

EDUCATION

San Jose State University **August 2022 - December 2026**
Bachelor of Science, Biomedical Engineering (Focus in Robotics) *San Jose, CA*

- **Coursework:** AI Control in Robotics, Engineering Design Methods, Linear Algebra, Differential Equations, Statistics in Engineering

TECHNICAL SKILLS

Languages: Python, C, Java, JSON, SQL, MATLAB, R
Robotics & Simulation: ROS 2, NVIDIA Isaac Sim/Isaac Lab, MuJoCo, PhysX, PyTorch, Digital Twins
Hardware & Embedded Systems: NVIDIA Jetson, Raspberry Pi, Arduino, EMG/IMU Sensors, Circuit Design, Laser Cutting, Bench Instruments, 7-Joint Robot Arms, Dextrous Robot Hands, Meta Quest 3, Vision Pro
Engineering Tools: SolidWorks, AutoCAD, Linux, Docker, Git
Verification & Validation: Hardware Validation, Functional Testing, Performance Testing, Requirements Traceability, ISO 13485